

- **VM (Virtual 8086 Mode, Bit 17):** When set (1), processor switches to Virtual 8086 mode.
 - **RF (Resume Flag, Bit 16):** Used with debug breakpoints to disable debug exceptions on the next instruction.
 - **IOPL (I/O Privilege Level, Bits 12-13):** 2-bit field (00 to 11) indicating minimum privilege required to execute I/O instructions.
 - **NT (Nested Task, Bit 14):** Set when a task is invoked via a **CALL** instruction with a Task State Segment (TSS).
-

4. Control Registers (CR0 - CR3)

- **CR0 (Machine Status Word Extended):**
 - **PE** (Bit 0 - Protection Enable): 1 = Protected Mode, 0 = Real Mode.
 - **PG** (Bit 31 - Paging Enable): 1 = Enable Paging MMU, 0 = Disable Paging.
 - **CR2 (Page Fault Linear Address):** Stores the 32-bit linear address that caused the last page fault exception.
 - **CR3 (Page Directory Base Register PDBR):** Stores the 32-bit physical base address of the top-level Page Directory.
-

2. Quick Recall Flow

80386 Registers -> 8 GPRs (EAX-ESP) -> 6 Segments (CS, DS, SS, ES, FS, GS) -> EFLAGS (VM, RF, IOPL, NT) -> Control (CR0: PE/PG, CR3: Page Directory)